

(Somaiya Vidyavihar University)

Department of Computer Engineering

Batch: C3 Roll No.: 16010123325

Experiment / Assignment / Tutorial No: 08

Title: Basic Reinforcement Learning using Q-Learning – Virtual Lab

Aim: To understand the fundamentals of Reinforcement Learning (RL) using the Virtual Labs simulator

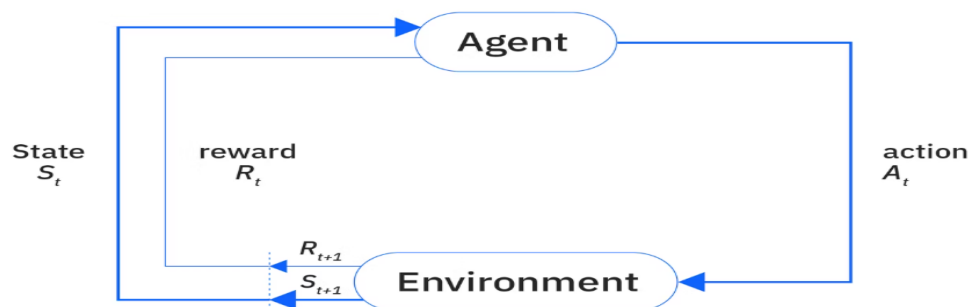
Expected Outcome of Experiment:

CO5: Gain a basic understanding of Reinforcement Learning (RL) principles and its potential applications.

Books/ Journals/ Websites referred:

1. Stuart Russell & Peter Norvig — *Artificial Intelligence: A Modern Approach*
2. Richard Sutton & Andrew Barto — *Reinforcement Learning: An Introduction*
3. Tom Mitchell — *Machine Learning*
4. <https://www.ibm.com/think/topics/reinforcement-learning>
5. <https://ai1-iiith.vlabs.ac.in/exp/q-learning/index.html>

Reinforcement Learning is a learning technique where an **agent learns by interacting with an environment through **trial and error** and improves its performance using **rewards and penalties**.**



Components of Reinforcement Learning

- ***Agent - Learner or decision maker***

Example:

Robot, game player, navigation system

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- **Environment - World in which the agent operates**

Example:

Gridworld simulation

- **State (S)- Current position or situation of agent**

Example:

Cell location inside grid

- **Action (A)- Possible moves agent can take**

Example:

1. Move Up
2. Move Down
3. Move Left
4. Move Right

- **Reward (R)- Feedback received after action**

Example:

Action Result	Reward
Correct move	+1
Wrong move	-1
Goal reached	+10

- **Policy (π) -Strategy followed by agent to choose actions**

Goal: Maximize total reward

Virtual Lab Support (Student Activity Steps)

Students should perform this experiment using:

<https://ai1-iiith.vlabs.ac.in/exp/q-learning/index.html>

Task

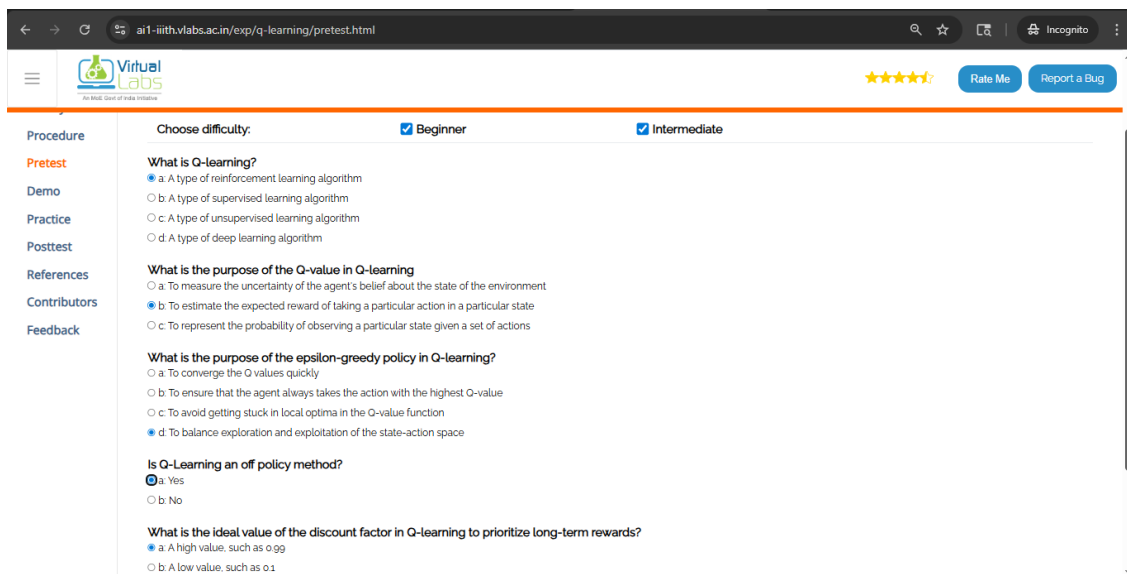
1. Identify: Start position, Goal position, Reward States, possible movement directions
2. Initialize the parameter – Learning Rate, Exploration rate etc.

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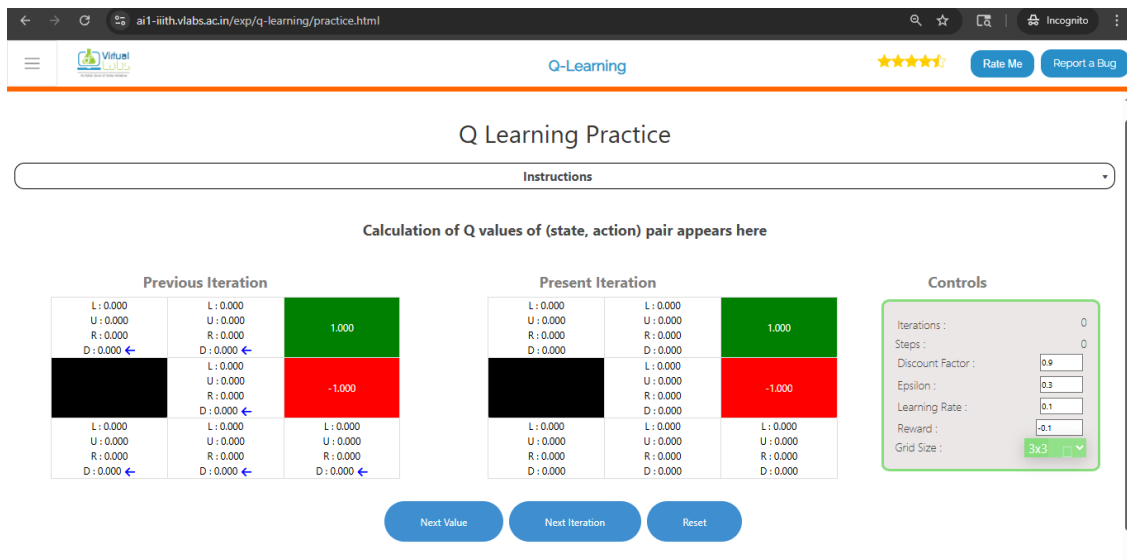
3. Monitor the Q-table updates
4. Repeat experiment by changing parameters

Results:



The screenshot shows the 'Pretest' section of the Virtual Labs interface. The left sidebar contains a menu with 'Pretest', 'Demo', 'Practice', 'Posttest', 'References', 'Contributors', and 'Feedback'. The main content area is titled 'Choose difficulty:' and has two tabs: 'Beginner' (selected) and 'Intermediate'. Below the tabs, there are several multiple-choice questions about Q-learning. The questions and their selected answers are:

- What is Q-learning?**
 - ☒ a A type of reinforcement learning algorithm
 - ☐ b A type of supervised learning algorithm
 - ☐ c A type of unsupervised learning algorithm
 - ☐ d A type of deep learning algorithm
- What is the purpose of the Q-value in Q-learning?**
 - ☐ a To measure the uncertainty of the agent's belief about the state of the environment
 - ☒ b To estimate the expected reward of taking a particular action in a particular state
 - ☐ c To represent the probability of observing a particular state given a set of actions
- What is the purpose of the epsilon-greedy policy in Q-learning?**
 - ☐ a To converge the Q values quickly
 - ☐ b To ensure that the agent always takes the action with the highest Q-value
 - ☐ c To avoid getting stuck in local optima in the Q-value function
 - ☒ d To balance exploration and exploitation of the state-action space
- Is Q-Learning an off policy method?**
 - ☒ a Yes
 - ☐ b No
- What is the ideal value of the discount factor in Q-learning to prioritize long-term rewards?**
 - ☒ a A high value, such as 0.99
 - ☐ b A low value, such as 0.1



The screenshot shows the 'Q Learning Practice' section of the Virtual Labs interface. The left sidebar contains a menu with 'Pretest', 'Demo', 'Practice', 'Posttest', 'References', 'Contributors', and 'Feedback'. The main content area is titled 'Q Learning Practice' and has a dropdown menu for 'Instructions'. Below the dropdown, there is a text box that says 'Calculation of Q values of (state, action) pair appears here'. The interface displays two 3x3 grids for 'Previous Iteration' and 'Present Iteration'. The 'Previous Iteration' grid shows Q-values for states L, U, R, D and actions L, U, R, D. The 'Present Iteration' grid shows the same Q-values after one iteration. The 'Controls' panel on the right allows users to set parameters for the simulation:

- Iterations: 0
- Steps: 0
- Discount Factor: 0.9
- Epsilon: 0.3
- Learning Rate: 0.1
- Reward: -0.1
- Grid Size: 3x3

At the bottom of the interface, there are three buttons: 'Next Value', 'Next Iteration', and 'Reset'.

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Q-Learning
★★★★★
Rate Me
Report a Bug

Q Learning Practice

Instructions

$$Q[(2,0),U] \leftarrow 0.000 + 0.100 * (-0.100 + 0.900 * 0.000 - (0.000)) = -0.010$$

Previous Iteration

L: 0.000 U: 0.000 R: 0.000 D: 0.000 ←	L: 0.000 U: 0.000 R: 0.000 D: 0.000 ←	1.000
	L: 0.000 U: 0.000 R: 0.000 D: 0.000 ←	-1.000
L: 0.000 U: 0.000 R: 0.000 D: 0.000 ←	L: 0.000 U: 0.000 R: 0.000 D: 0.000 ←	L: 0.000 U: 0.000 R: 0.000 D: 0.000 ←

Present Iteration

L: 0.000 U: 0.000 R: 0.000 D: 0.000	L: 0.000 U: 0.000 R: 0.000 D: 0.000	1.000
	L: 0.000 U: 0.000 R: 0.000 D: 0.000	-1.000
L: -0.010 U: -0.010 R: 0.000 D: 0.000	L: 0.000 U: 0.000 R: 0.000 D: 0.000	L: 0.000 U: 0.000 R: 0.000 D: 0.000

Controls

Iterations: 0

Steps: 2

Discount Factor: 0.9

Epsilon: 0.3

Learning Rate: 0.1

Reward: -0.1

Grid Size: 3x3

Next Value
Next Iteration
Reset



Q-Learning
★★★★★
Rate Me
Report a Bug

Q Learning Practice

Instructions

$$Q[(2,0),R] \leftarrow -0.042 + 0.100 * (-0.100 + 0.900 * -0.010 - (-0.042)) = -0.049$$

Previous Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000 →	L: -0.010 U: 0.000 R: 0.000 D: 0.000 ↑	1.000
	L: 0.000 U: -0.010 R: 0.000 D: -0.011 ←	-1.000
L: -0.029 U: -0.038 R: -0.042 D: -0.030 ←	L: -0.022 U: -0.019 R: -0.019 D: -0.010 ↓	L: -0.010 U: 0.000 R: 0.000 D: -0.010 ↑

Present Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000	L: -0.010 U: 0.000 R: 0.000 D: 0.000	1.000
	L: 0.000 U: -0.010 R: 0.000 D: -0.011	-1.000
L: -0.048 U: -0.048 R: -0.049 D: -0.049	L: -0.022 U: -0.019 R: -0.019 D: -0.010	L: -0.010 U: 0.000 R: 0.000 D: -0.010

Controls

Iterations: 3

Steps: 6

Discount Factor: 0.9

Epsilon: 0.45

Learning Rate: 0.1

Reward: -0.1

Grid Size: 3x3

Next Value
Next Iteration
Reset

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Q-Learning

Q Learning Practice

Instructions

$$Q[(2,1),D] \leftarrow -0.020 + 0.100 * (-0.100 + 0.900 * -0.020 - (-0.020)) = -0.030$$

Previous Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000 →	L: -0.010 U: 0.000 R: 0.000 D: 0.000 ↑	1.000
	L: -0.010 U: -0.010 R: -0.100 D: -0.011 ←	-1.000
L: -0.058 U: -0.058 R: -0.055 D: -0.049 ↓	L: -0.022 U: -0.027 R: -0.100 D: -0.020 ↓	L: -0.010 U: -0.100 R: 0.000 D: -0.010 →

Present Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000	L: -0.010 U: 0.000 R: 0.000 D: 0.000	1.000
	L: -0.010 U: -0.010 R: -0.100 D: -0.011 ←	-1.000
L: -0.058 U: -0.058 R: -0.062 D: -0.058	L: -0.022 U: -0.027 R: -0.027 D: -0.030	L: -0.010 U: -0.100 R: 0.000 D: -0.010

Controls

Iterations: 5

Steps: 3

Discount Factor: 0.9

Epsilon: 0.45

Learning Rate: 0.1

Reward: -0.1

Grid Size: 3x3

Next Value Next Iteration Reset

Q-Learning

Q Learning Practice

Instructions

$$Q[(2,0),L] \leftarrow -0.058 + 0.100 * (-0.100 + 0.900 * -0.058 - (-0.058)) = -0.067$$

Previous Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000 →	L: -0.010 U: 0.000 R: 0.000 D: 0.000 ↑	1.000
	L: -0.010 U: -0.010 R: -0.100 D: -0.011 ←	-1.000
L: -0.058 U: -0.058 R: -0.055 D: -0.049 ↓	L: -0.022 U: -0.027 R: -0.100 D: -0.020 ↓	L: -0.010 U: -0.100 R: 0.000 D: -0.010 →

Present Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000	L: -0.010 U: 0.000 R: 0.000 D: 0.000	1.000
	L: -0.010 U: -0.010 R: -0.100 D: -0.011 ←	-1.000
L: -0.067 U: -0.067 R: -0.062 D: -0.058	L: -0.035 U: -0.027 R: -0.027 D: -0.039	L: -0.010 U: -0.100 R: 0.000 D: -0.010

Controls

Iterations: 5

Steps: 7

Discount Factor: 0.9

Epsilon: 0.45

Learning Rate: 0.1

Reward: -0.1

Grid Size: 3x3

Next Value Next Iteration Reset

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Q-Learning

Q Learning Practice

Instructions

$$Q[2,1,U] \leftarrow -0.027 + 0.100 * (-0.100 + 0.900 * -0.010 - (-0.027)) = -0.035$$

Previous Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000 →	L: -0.010 U: 0.000 R: 0.000 D: 0.000 ↑	1.000
	L: -0.010 U: -0.010 R: -0.100 D: -0.011 ←	-1.000
L: -0.076 U: -0.067 R: -0.062 D: -0.077 →	L: -0.035 U: -0.027 R: -0.027 D: -0.039 ↑	L: -0.010 U: -0.100 R: 0.000 D: -0.010 →

Present Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000	L: -0.010 U: 0.000 R: 0.000 D: 0.000	1.000
	L: -0.010 U: -0.010 R: -0.100 D: -0.011	-1.000
L: -0.076 U: -0.076 R: -0.068 D: -0.077	L: -0.035 U: -0.035 R: -0.027 D: -0.039	L: -0.010 U: -0.100 R: 0.000 D: -0.010

Controls

Iterations: 6
Steps: 3
Discount Factor: 0.9
Epsilon: 0.45
Learning Rate: 0.1
Reward: -0.1
Grid Size: 3x3

Next Value Next Iteration Reset

Q-Learning

Q Learning Practice

Instructions

$$Q[1,1,U] \leftarrow -0.019 + 0.100 * (-0.100 + 0.900 * 0.080 - (-0.019)) = -0.020$$

Previous Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000 →	L: -0.010 U: -0.010 R: 0.080 D: 0.000 →	1.000
	L: -0.020 U: -0.019 R: -0.190 D: -0.011 ↓	-1.000
L: -0.102 U: -0.085 R: -0.089 D: -0.103 ↑	L: -0.048 U: -0.043 R: -0.047 D: -0.048 ↑	L: -0.022 U: -0.190 R: -0.010 D: -0.010 →

Present Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000	L: -0.010 U: -0.010 R: 0.080 D: 0.000	1.000
	L: -0.020 U: -0.020 R: -0.190 D: -0.011	-1.000
L: -0.102 U: -0.085 R: -0.094 D: -0.103	L: -0.048 U: -0.049 R: -0.047 D: -0.048	L: -0.022 U: -0.190 R: -0.010 D: -0.010

Controls

Iterations: 10
Steps: 3
Discount Factor: 0.9
Epsilon: 0.45
Learning Rate: 0.1
Reward: -0.1
Grid Size: 3x3

Next Value Next Iteration Reset

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Q-Learning

Q Learning Practice

Instructions

$$Q[(2,0),U] \leftarrow -0.094 + 0.100 * (-0.100 + 0.900 * -0.094 - (-0.094)) = -0.103$$

Previous Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000 →	L: -0.010 U: -0.010 R: 0.152 D: 0.000 →	1.000
	L: -0.020 U: -0.020 R: -0.271 D: -0.011 ↓	-1.000
L: -0.102 U: -0.094 R: -0.099 D: -0.103 ↑	L: -0.048 U: -0.055 R: -0.047 D: -0.048 →	L: -0.022 U: -0.190 R: -0.010 D: -0.010 →

Present Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000	L: -0.010 U: -0.010 R: 0.152 D: 0.000	1.000
	L: -0.020 U: -0.020 R: -0.271 D: -0.011	-1.000
L: -0.102 U: -0.103 R: -0.099 D: -0.103	L: -0.048 U: -0.055 R: -0.047 D: -0.048	L: -0.022 U: -0.190 R: -0.010 D: -0.010

Controls

Iterations: 12

Steps: 1

Discount Factor: 0.9

Epsilon: 0.45

Learning Rate: 0.1

Reward: -0.1

Grid Size: 3x3

Next Value Next Iteration Reset

Q-Learning

Q Learning Practice

Instructions

$$Q[(2,1),R] \leftarrow -0.075 + 0.100 * (-0.100 + 0.900 * -0.039 - (-0.075)) = -0.081$$

Previous Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000 →	L: -0.010 U: -0.010 R: 0.152 D: 0.000 →	1.000
	L: -0.030 U: -0.020 R: -0.344 D: -0.024 ↑	-1.000
L: -0.127 U: -0.137 R: -0.128 D: -0.129 ←	L: -0.077 U: -0.072 R: -0.075 D: -0.067 ↓	L: -0.048 U: -0.344 R: -0.039 D: -0.039 →

Present Iteration

L: -0.019 U: -0.010 R: 0.000 D: 0.000	L: -0.010 U: -0.010 R: 0.152 D: 0.000	1.000
	L: -0.030 U: -0.020 R: -0.344 D: -0.024	-1.000
L: -0.136 U: -0.145 R: -0.132 D: -0.129	L: -0.077 U: -0.072 R: -0.081 D: -0.067	L: -0.048 U: -0.344 R: -0.039 D: -0.039

Controls

Iterations: 18

Steps: 4

Discount Factor: 0.9

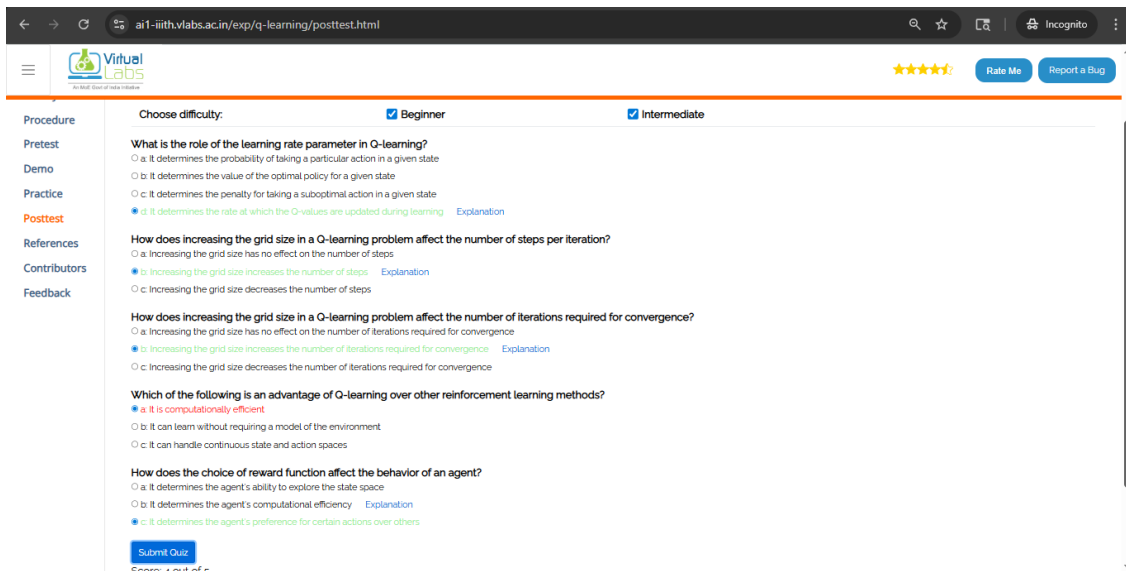
Epsilon: 0.45

Learning Rate: 0.1

Reward: -0.1

Grid Size: 3x3

Next Value Next Iteration Reset



ai1-iiith.vlabs.ac.in/exp/q-learning/posttest.html

Virtual Labs

Choose difficulty: ☒ Beginner ☒ Intermediate

Procedure
Pretest
Demo
Practice
Posttest
References
Contributors
Feedback

What is the role of the learning rate parameter in Q-learning?

- ☐ a. It determines the probability of taking a particular action in a given state
- ☐ b. It determines the value of the optimal policy for a given state
- ☐ c. It determines the penalty for taking a suboptimal action in a given state
- ☒ d. It determines the rate at which the Q-values are updated during learning [Explanation](#)

How does increasing the grid size in a Q-learning problem affect the number of steps per iteration?

- ☐ a. Increasing the grid size has no effect on the number of steps
- ☒ b. Increasing the grid size increases the number of steps [Explanation](#)
- ☐ c. Increasing the grid size decreases the number of steps

How does increasing the grid size in a Q-learning problem affect the number of iterations required for convergence?

- ☐ a. Increasing the grid size has no effect on the number of iterations required for convergence
- ☒ b. Increasing the grid size increases the number of iterations required for convergence [Explanation](#)
- ☐ c. Increasing the grid size decreases the number of iterations required for convergence

Which of the following is an advantage of Q-learning over other reinforcement learning methods?

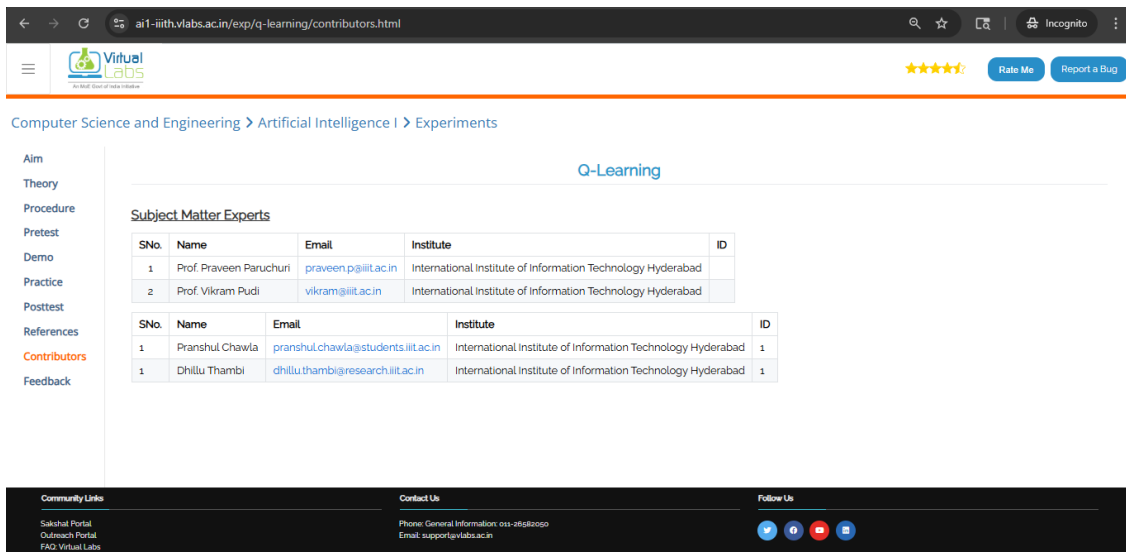
- ☒ a. It is computationally efficient
- ☐ b. It can learn without requiring a model of the environment
- ☐ c. It can handle continuous state and action spaces

How does the choice of reward function affect the behavior of an agent?

- ☐ a. It determines the agent's ability to explore the state space
- ☐ b. It determines the agent's computational efficiency [Explanation](#)
- ☒ c. It determines the agent's preference for certain actions over others

[Submit Quiz](#)

Score: 4 out of 5



ai1-iiith.vlabs.ac.in/exp/q-learning/contributors.html

Virtual Labs

Computer Science and Engineering > Artificial Intelligence > Experiments

Aim
Theory
Procedure
Pretest
Demo
Practice
Posttest
References
Contributors
Feedback

Q-Learning

Subject Matter Experts

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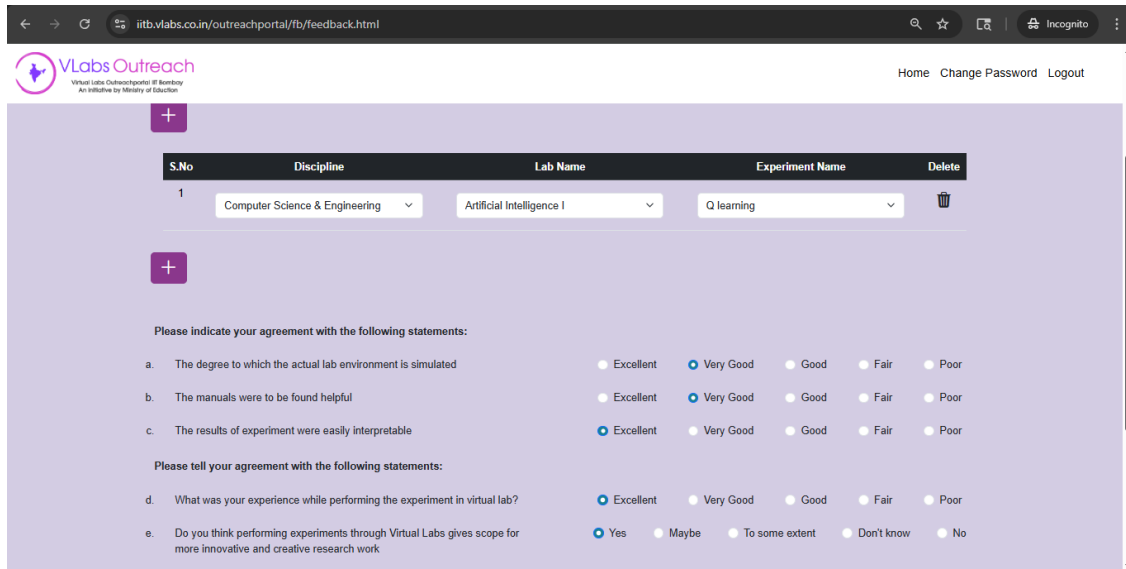
Community Links
[Syllabus Portal](#)
[Outreach Portal](#)
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 Phone: General Information: 020-26120200
 Email: support@iiit.ac.in

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The screenshot shows a web browser window with the URL `iitb.vlabs.co.in/outreachportal/fb/feedback.html`. The page is titled "VLabs Outreach" and includes a navigation bar with "Home", "Change Password", and "Logout". Below the header, there is a table with the following columns: "S.No", "Discipline", "Lab Name", "Experiment Name", and "Delete". The table contains one row with the following data: "1", "Computer Science & Engineering", "Artificial Intelligence I", "Q learning", and a delete icon. Below the table, there is a section for feedback. It starts with a heading "Please indicate your agreement with the following statements:" followed by three statements (a, b, c) and their corresponding radio button options. Statement (a) is "The degree to which the actual lab environment is simulated" with options: Excellent, Very Good (selected), Good, Fair, Poor. Statement (b) is "The manuals were to be found helpful" with options: Excellent, Very Good (selected), Good, Fair, Poor. Statement (c) is "The results of experiment were easily interpretable" with options: Excellent (selected), Very Good, Good, Fair, Poor. Below this, there is another heading "Please tell your agreement with the following statements:" followed by two statements (d, e) and their corresponding radio button options. Statement (d) is "What was your experience while performing the experiment in virtual lab?" with options: Excellent (selected), Very Good, Good, Fair, Poor. Statement (e) is "Do you think performing experiments through Virtual Labs gives scope for more innovative and creative research work" with options: Yes (selected), Maybe, To some extent, Don't know, No.

Conclusion:

The experiment successfully demonstrated the working of **Reinforcement Learning using Q-Learning** through a grid-based virtual environment. The agent learned to navigate from the initial state to the goal state by interacting with the environment and updating Q-values using rewards and penalties. It was observed that over multiple iterations, the agent gradually improved its policy and converged to the optimal path that maximizes cumulative reward while avoiding negative reward states. The effect of parameters such as learning rate, discount factor, and epsilon on learning behavior was also understood. This experiment highlights how Q-learning enables model-free learning and decision-making in dynamic environments.

Post Lab :

1. Define **Reinforcement Learning** and explain how it differs from supervised learning.

Reinforcement Learning (RL):

A machine learning technique where an agent learns by interacting with an environment through trial and error, receiving rewards or penalties, and aims to maximize cumulative reward.

Difference from Supervised Learning:

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Reinforcement Learning	Supervised Learning
Learns via interaction with environment	Learns from labeled dataset
No predefined correct output	Has correct input-output pairs
Feedback is reward/penalty	Feedback is direct error/loss
Focus on sequential decisions	Focus on prediction

2. What is a **policy** in Reinforcement Learning?

A **policy** (π) is a strategy or rule followed by an agent to decide which action to take in a given state. It maps states to actions and determines the behavior of the agent. The objective of the policy is to maximize the total cumulative reward over time.